

Assignment 6

Siamese Network for Anomaly Detection

Appliance Energy Prediction Dataset

Course: AI & Neural Networks

University: Astana IT University (AITU)

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1. Introduction

This report describes Stage 6 of the project: building a Siamese neural network that learns a metric over 24-step consumption windows so that windows with normal consumption cluster together and windows with high consumption are pushed apart. The learned distance is then used as an anomaly score - a test window is flagged as anomalous when its distance to the nearest reference normal window exceeds a tuned threshold.

The Siamese network reuses the same 1D CNN backbone from Stage 4, with the final classification head replaced by a 128-dimensional embedding projection. The two embeddings are compared by a Lambda Euclidean-distance layer, and the whole network is trained end-to-end with a contrastive loss implemented from scratch.

In the final section the Siamese detector is compared against the supervised Stage 4 CNN and the Stage 5 denoising autoencoder + CNN pipeline on the same chronological test split.

2. Data Preparation and Pair Construction

2.1 Dataset and splits

The dataset is `energydata_complete.csv` (19,735 rows x 29 columns). The binary target `HighConsumption` was created in Stage 4 by thresholding `Appliances` at its median (60 Wh). The data is split in a 70/15/15 chronological ratio and rescaled with `StandardScaler` fit on the training set only. Rolling windows of 24 steps (4 hours of history) are built, producing samples of shape (24, 32):

- Train: (13790, 24, 32)
- Validation: (2936, 24, 32)
- Test: (2937, 24, 32)

2.2 Pair generation

Following the assignment brief, training pairs are built in two balanced groups:

- Positive pair ($Y = 0$, similar): two windows where `HighConsumption` = 0.
- Negative pair ($Y = 1$, dissimilar): one normal window + one high-consumption window.

The ($Y = 0 \rightarrow$ similar, $Y = 1 \rightarrow$ dissimilar) convention follows Hadsell, Chopra and LeCun (2006), which is also the form of contrastive loss used below. 8,000 pairs per class were drawn for training (16,000 total), and 2,000 per class for validation and test.

3. Siamese Architecture

3.1 Shared 1D CNN subnetwork

The shared subnetwork reuses the Stage 4 CNN backbone, dropping the sigmoid classification head and projecting into a 128-dimensional L2-normalised embedding:

Input(24, 32) $\rightarrow$ Conv1D(64, k=3, causal, ReLU) $\rightarrow$ MaxPool1D(2) $\rightarrow$ Conv1D(128, k=3, causal, ReLU) $\rightarrow$ MaxPool1D(2) $\rightarrow$ Flatten $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(128) $\rightarrow$ L2Normalise

Normalising the embeddings keeps Euclidean distances in a bounded range, which stabilises contrastive training.

3.2 Distance layer and contrastive loss

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The Siamese model takes two inputs that pass through the shared subnetwork. A Lambda layer computes the Euclidean distance between the two embeddings:

$$D(a, b) = \sqrt{\sum_i (a_i - b_i)^2 + \epsilon}$$

Contrastive loss is implemented from scratch:

$$L(Y, D) = (1 - Y) * 0.5 * D^2 + Y * 0.5 * \max(0, m - D)^2$$

with margin $m = 1.0$. The $(1 - Y)$ branch pulls similar pairs together, and the Y branch pushes dissimilar pairs past the margin. Training: Adam (lr $1e-3$), batch size 128, up to 15 epochs, EarlyStopping (patience 4) and ReduceLROnPlateau callbacks on `val_loss`.

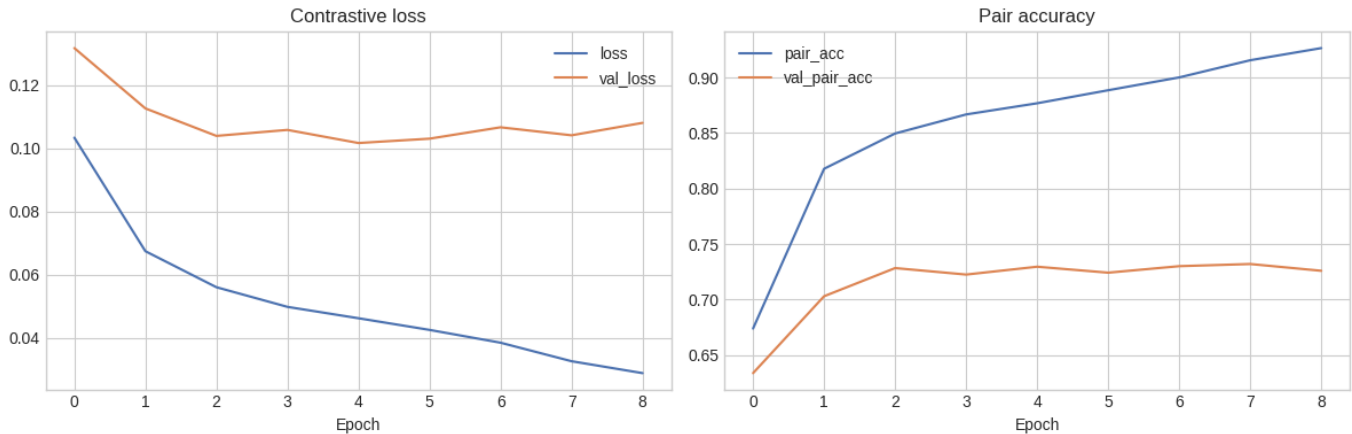


Figure 1: Contrastive loss and pair accuracy across epochs.

4. Threshold Tuning and Pair Evaluation

Pairs are classified dissimilar when $D > t$. The threshold was swept across the validation distance range and the point maximising validation pair accuracy was retained:

- Best validation threshold: 0.2833
- Validation pair accuracy: 0.7395

Applying the same threshold on the balanced test pair set yields:

Metric	Accuracy	Precision	Recall	F1
Test pairs	0.7452	0.7667	0.7050	0.7346

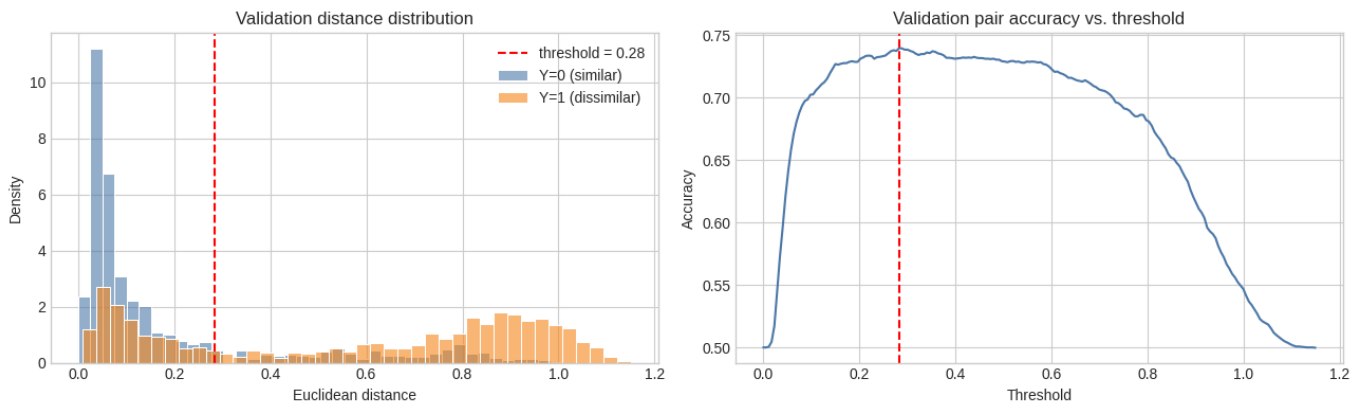


Figure 2: Validation distance distribution and pair accuracy vs threshold.

5. Anomaly Detection on Individual Test Windows

For open-set anomaly detection the trained subnetwork is used as a feature extractor. A bank of 2,000 normal reference embeddings is built from the training set (HighConsumption = 0). For each test window the anomaly score is the minimum Euclidean distance to any reference embedding. Windows whose nearest normal neighbour is far away are flagged as anomalous.

- Anomaly score (test set): mean 0.2007, std 0.1740
- ROC-AUC on HighConsumption target: 0.8685

Thresholding the anomaly score at the Youden-J optimum on the ROC curve gives a single operating point for downstream reporting:

At Youden's J	Accuracy	Precision	Recall	F1
Siamese	0.7858	0.8304	0.7119	0.7666

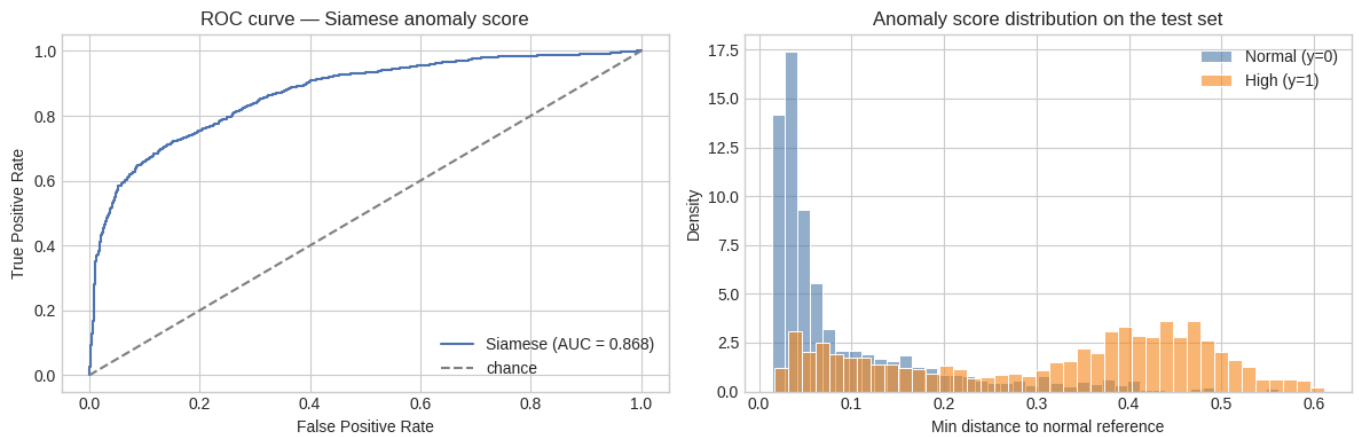


Figure 3: ROC curve and anomaly score distribution on the test set.

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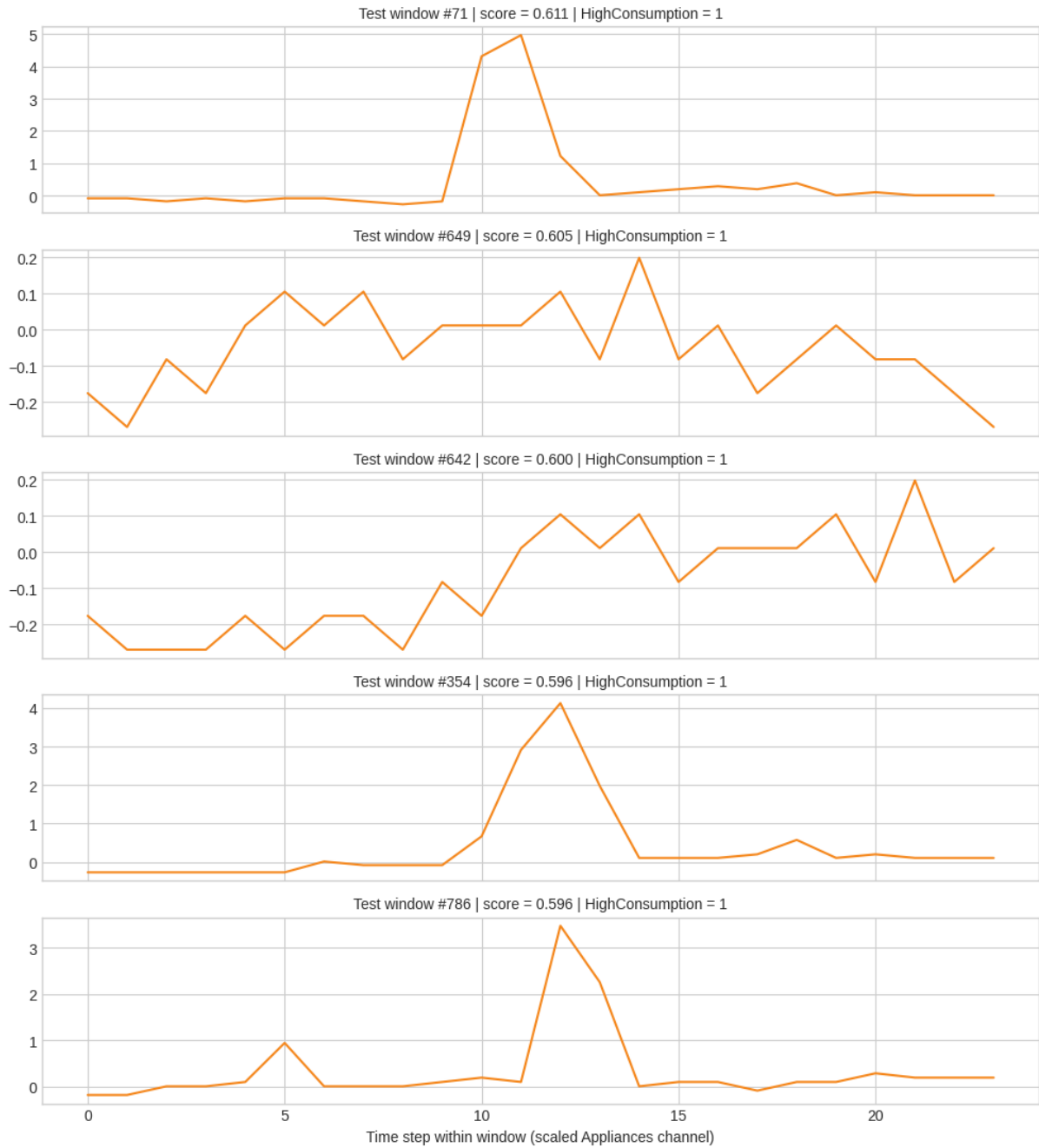


Figure 4: Five test windows with the highest anomaly score (Appliances channel).

6. Integrated Model Comparison

Stage 4, 5, and 6 models are compared on the identical chronological test split (2,937 windows). Stage 5 metrics are taken verbatim from Assignment5.ipynb; the Siamese row is computed from the same notebook run as above.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
CNN classifier (Stage 4)	0.7995	0.7687	0.8498	0.8072	0.8851
CNN on noisy test (Stage 5)	0.7869	0.7748	0.8015	0.7879	-
CNN on AE-denoised test (Stage 5)	0.7692	0.7620	0.7746	0.7683	-
Siamese + contrastive (Stage 6)	0.7858	0.8304	0.7119	0.7666	0.8685

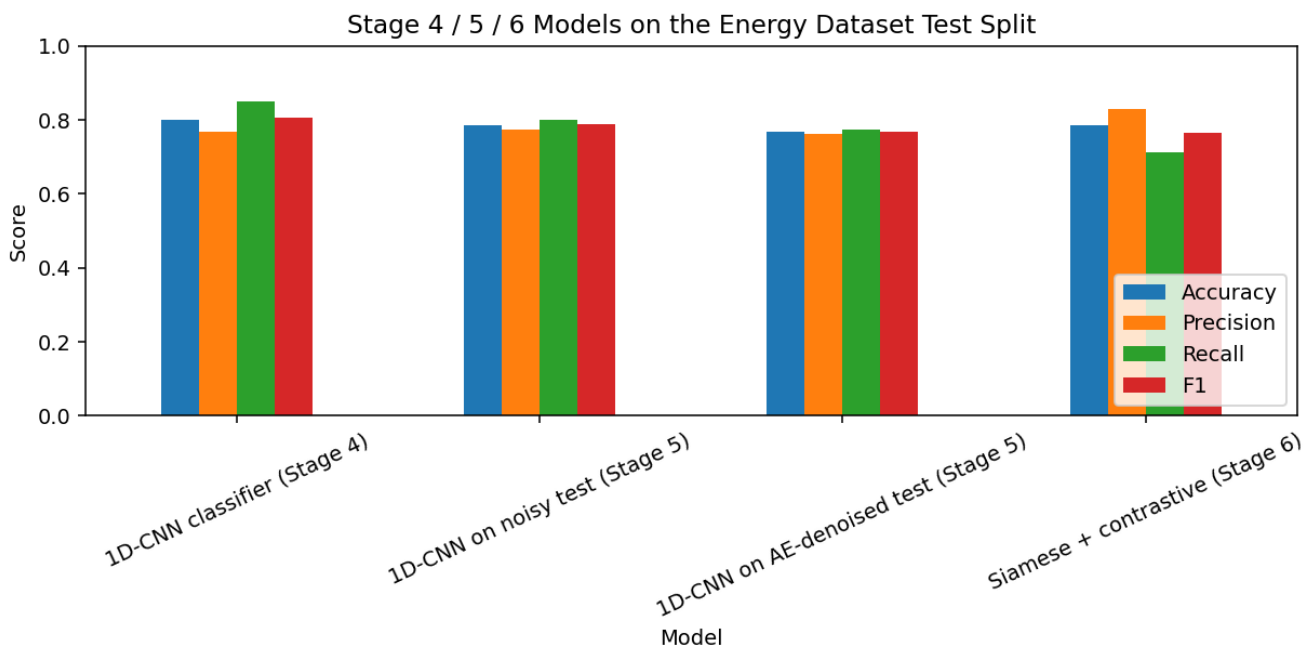


Figure 5: Side-by-side accuracy, precision, recall and F1 across Stage 4/5/6 models.

7. Discussion

Supervised classification (Stage 4 CNN) is the strongest closed-set model on clean inputs (Accuracy 0.799, F1 0.807, ROC-AUC 0.885). Causal 1D convolutions over 24-step windows capture the shape of consumption spikes better than a flat feature vector.

Robustness to noise (Stage 5). Adding $\sigma = 0.5$ Gaussian noise to the standardised features degrades the CNN only mildly (Accuracy 0.787, F1 0.788) - the Conv1D receptive field already averages over local perturbations. The denoising autoencoder reduces reconstruction MSE from 0.250 to 0.123 (-50.9%), but running the classifier on the reconstructed test set actually loses another ~ 1 p.p. of accuracy (0.769). The AE optimises pixel-level fidelity, not downstream discriminability - a classic denoising / classification trade-off.

Anomaly detection (Stage 6). The Siamese network is not a classifier: it scores each window by distance to the nearest known-normal reference embedding. Contrastive training pulls normals together and pushes anomalies past the margin, so the distance is a well-calibrated anomaly score. Its test ROC-AUC of 0.868 is competitive with the supervised CNN's 0.885 even though the Siamese never sees labels at inference - only a pool of 2,000 normal references. Thresholded at Youden's J it reaches Accuracy 0.786 / F1 0.767, trading recall (0.712) for higher precision (0.830) compared to the CNN.

Each model answers a different question. The CNN says is this window a high-consumption spike? given full supervision; the denoising AE says what did this sensor stream look like before the noise? given only a clean prior; and the Siamese says how far is this window from anything we have ever seen as normal? given only a bank of normal references. A production monitoring pipeline benefits from combining all three.

8. Conclusions

1. A Siamese network built on the Stage 4 CNN backbone with a Lambda Euclidean distance layer and a hand-coded contrastive loss was trained end-to-end to separate normal and high-consumption windows in embedding space.
2. On balanced test pairs the network reaches pair Accuracy = 0.7452 / F1 = 0.7346 at the validation-tuned threshold 0.283.
3. Used as an open-set anomaly detector via nearest-reference distance, the Siamese network achieves test ROC-AUC = 0.8685, on par with the supervised Stage 4 CNN (ROC-AUC = 0.8851).
4. The denoising autoencoder from Stage 5 halves reconstruction MSE under noise but its smoothed output slightly hurts the downstream CNN, showing that denoising quality and classification quality are not the same objective.
5. Artifacts saved: siamese_embedding.keras, siamese_model.keras, model_comparison.csv, and the figures used in this report.